

Educational Background

09/2022 - Present B.Eng. in Electronic Information Engineering Guilin University of Electronic Technology

Core Course Modules: Communication, Electronic Circuits, Analog Electronic Technology, Digital Logic, Signal and System Analysis, Fundamentals of Circuit Analysis.

Research Experience

04/2025 Measurement and Application of FMCW Radio Altimeter

Project Description: Achieved a 20G FMCW signal output using Xilinx Zynq7020 as the control core. AD9361 is controlled by the FPGA to transmit RF and echo signals. Filter parameters are configured through the ADI software-defined radio platform. After filtering, signals undergo FFT to obtain the distance.

Responsibilities: 1. Develop the Zedboard AD9361 core board, programme AD9361, configure filter parameters, design the FFT and DMA routines, and generate FMCW signals through AD9361. 2. Design and assemble the host computer. 3. Transmit altitude data using Lora. 4. Design and fabricate the radar antenna structure using Fusion 360 and 3D printing.

08/2025 A Multi-Scenario Weak Magnetic Field Measurement System Based on TMR Sensors

Project Description: Construct a weak magnetic field measurement system with two microcontrollers (STM32C8T6). It measures the magnetic field using TMR2584 sensors, with a measurement range of <200 Oe and an accuracy of ~8.0%. The measured signal is transmitted to the host computer via the SPI interface with a response time <5 ms. A power supply board is designed using LT3045 and LT3015 linear regulator ICs to provide positive and negative voltages, respectively, with RMS noise <0.8 μVrms (from 10 Hz to 100 kHz). The output current is up to 1 A with a voltage range of 3.3 - 20 V.

Responsibilities: 1. Overall project design and implementation, detecting and transmitting magnetic field signals and Develop the interface for data display and modification. 2. Development of the host computer. 3. Design the power supply board.

Professional Skills

College English Test Band 4:	Capable of reading technical documents and data manuals in English.
Electronic Testing Instruments:	Multimeters, oscilloscopes, signal generators, and power supplies, etc.
Embedded System Development:	KEIL 5 for developing STM32 and 51 microcontrollers, debugging and developing in C.
Circuit Simulators:	Multisim & Proteus.
Communication Protocols:	UART, I2C, and SPI.
Other Development Tools:	Quartus for FPGA development. MATLAB for numerical simulation.

Honours & Awards

2024	Guangxi University Student Electronic Design Competition	Provincial Second Prize
2024	"Tongda Cup" ADI Software Radio Competition	National Second Prize
2025	"Bank of China Cup" China University Students International Innovation Competition	Provincial Third Prize

Publications

- Z. Chen**, X. Ding, W. Wan, Y. Liu, C. Cao, and T. Liu, "Measurement and Application of Radio Altimeter Based on FMCW", in *Proceedings of 5th International Conference on Consumer Electronics and Computer Engineering (ICCECE)*, Dongguan, China, 2025, pp. 511 - 515.
- Y. Jiang, S. Li, **Z. Chen**, X. Ding, J. Cao, and T. Liu, "Research on Simulation of Constant Difference Frequency Radio Altimeter Based on FMCW", in *Proceedings of 5th International Conference on Intelligent Communications and Computing (ICICC)*, Nanjing, China, 2025, pp. 132 – 136.
- Z. Chen** (First Designer). LORA Height Table Data Reception Platform [Computer Software Copyright]. Registration No.2025SR0803908. China, 2025.